

AdaptiveLogSoftmaxWithLoss#

<https://pytorch.org/docs/stable/generated/torch.nn.AdaptiveLogSoftmaxWithLoss>

Description:

Efficient softmax approximation. As described in Efficient softmax approximation for GPUs by Edouard Grave, Armand Joulin, Moustapha Cissé, David Grangier, and Hervé Jégou. Adaptive softmax is an approximate strategy for training models with large output spaces. It is most effective when the label distribution is highly imbalanced, for example in natural language modelling, where the word frequency distribution approximately follows the Zipf's law. Adaptive softmax partitions the labels into several clusters, according to their frequency. These clusters may contain different number of targets each. Additionally, clusters containing less frequent labels assign lower dimensional embeddings to those labels, which speeds up the computation. For each minibatch, only clusters for which at least one target is present are evaluated. The idea is that the clusters which are accessed frequently (like the first one, containing most frequent labels), should also be cheap to compute – that is, contain a small number of assigned labels.

Function Signature

```
class torch.nn.AdaptiveLogSoftmaxWithLoss(in_features,  
n_classes, cutoffs, div_value=4.0, head_bias=False,  
device=None, dtype=None)[source]#
```

Parameters

Returns

Return type

Shape:

Returns:

output (Tensor)

Notes & Warnings

WARNING

Warning Labels passed as inputs to this module should be sorted according to their frequency. This means that the most frequent label should be represented by the index 0, and the least frequent label should be represented by the index $n_classes - 1$.

NOTE

Note This module returns a NamedTuple with output and loss fields. See further documentation for details.

NOTE

Note To compute log-probabilities for all classes, the `log_prob` method can be used.

Detailed Documentation

Documentation

Efficient softmax approximation.

As described in Efficient softmax approximation for GPUs by Edouard Grave, Armand Joulin, Moustapha Cissé, David Grangier, and Hervé Jégou.

Adaptive softmax is an approximate strategy for training models with large output spaces. It is most effective when the label distribution is highly imbalanced, for example

in natural language modelling, where the word frequency distribution approximately follows the Zipf's law.

Adaptive softmax partitions the labels into several clusters, according to their frequency. These clusters may contain different number of targets each. Additionally, clusters containing less frequent labels assign lower dimensional embeddings to those labels, which speeds up the computation. For each minibatch, only clusters for which at least one target is present are evaluated.

The idea is that the clusters which are accessed frequently (like the first one, containing most frequent labels), should also be cheap to compute – that is, contain a small number of assigned labels.

We highly recommend taking a look at the original paper for more details.

cutoffs should be an ordered Sequence of integers sorted in the increasing order. It controls number of clusters and the partitioning of targets into clusters. For example setting cutoffs = [10, 100, 1000] means that first 10 targets will be assigned to the 'head' of the adaptive softmax, targets 11, 12, ..., 100 will be assigned to the first cluster, and targets 101, 102, ..., 1000 will be assigned to the second cluster, while targets 1001, 1002, ..., n_classes - 1 will be assigned to the last, third cluster.

div_value is used to compute the size of each additional cluster, which is given as $\left\lfloor \frac{\text{in_features}}{\text{div_value}^{\text{idx}}} \right\rfloor$, where idx is the cluster index (with clusters for less frequent words having larger indices, and indices starting from 1).

head_bias if set to True, adds a bias term to the 'head' of the adaptive softmax. See paper for details. Set to False in the official implementation.

Labels passed as inputs to this module should be sorted according to their frequency.

This means that the most frequent label should be represented by the index 0, and the least frequent label should be represented by the index $n_classes - 1$.

This module returns a `NamedTuple` with `output` and `loss` fields. See further documentation for details.

To compute log-probabilities for all classes, the `log_prob` method can be used.

`in_features` (int) – Number of features in the input tensor

`n_classes` (int) – Number of classes in the dataset

`cutoffs` (Sequence) – Cutoffs used to assign targets to their buckets

`div_value` (float, optional) – value used as an exponent to compute sizes of the clusters.
Default: 4.0

`head_bias` (bool, optional) – If True, adds a bias term to the ‘head’ of the adaptive softmax. Default: False

`output` is a Tensor of size N containing computed target log probabilities for each example

`loss` is a Scalar representing the computed negative log likelihood loss

`NamedTuple` with `output` and `loss` fields

`input`: $(N, in_features)$ or $(N, in_features)$ or $(in_features)$

`target`: $(N)(N)(N)$ or $()()$ where each value satisfies $0 \leq target[i] \leq n_classes$

output1: (N)(N)(N) or ()()

Runs the forward pass.

Compute log probabilities for all $n_classes$.

input (Tensor) – a minibatch of examples

log-probabilities of for each class c in range $0 \leq c \leq n_classes$, where $n_classes$ is a parameter passed to AdaptiveLogSoftmaxWithLoss constructor.

Input: (N,in_features)(N, in_features)(N,in_features)

Output: (N,n_classes)(N, n_classes)(N,n_classes)

Return the class with the highest probability for each example in the input minibatch.

This is equivalent to `self.log_prob(input).argmax(dim=1)`, but is more efficient in some cases.

input (Tensor) – a minibatch of examples

a class with the highest probability for each example

Input: (N,in_features)(N, in_features)(N,in_features)

Resets parameters based on their initialization used in `__init__`.